



Debugging Lecture I

ECEN 3314, Fall 2026

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Outline

In a few mini-lectures, I would like to teach you:

1. Common Circuit mistakes (Source: A six-part series posted by TI Precision Lab)
2. Global debugging strategies and mindset (based on my NSF EDU research, 2023-2026)
3. AI-Assisted debugging tips (based on my NSF CAREER research: 2026-2031)

We would happily be your **debugging guy!**

- Share with us your headaches and wish-lists at john.hu@okstate.edu.



Common Circuit Mistake #1: Connectivity

A circuit that works on paper does not work on a board. Why?

Common Mistake # 1: **connectivity error**

- You may have **missed** some connections
- You may have connected something **wrong**: unexpected open circuit, short circuit, or just wrong connections

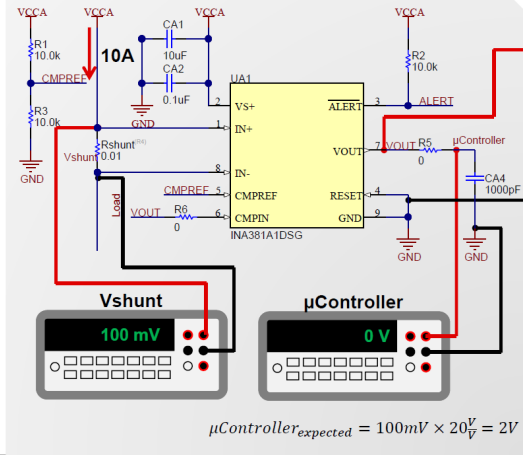
How to spot those mistakes?

- Visual inspection: female students may have an edge ([Ash and Hu 2025](#)), but it may depend on the type of circuit and bug; it may also due to their patience.
- Other methods to be discussed later



Example 1: TI Current Sense Amplifier

Conditions



Let's use a TI current sensing chip as an example. We want to measure an unknown current I . The chip lets I flow through a known resistance, R_{shunt} .

$$\therefore V_{shunt} = IR_{shunt} \rightarrow I = \frac{V_{shunt}}{R_{shunt}} \quad (1)$$

In this example, $I = 10A$. Since we used a $R_{shunt} = 10m\Omega$ on the board, we measured $100mV$ across R_{shunt} .

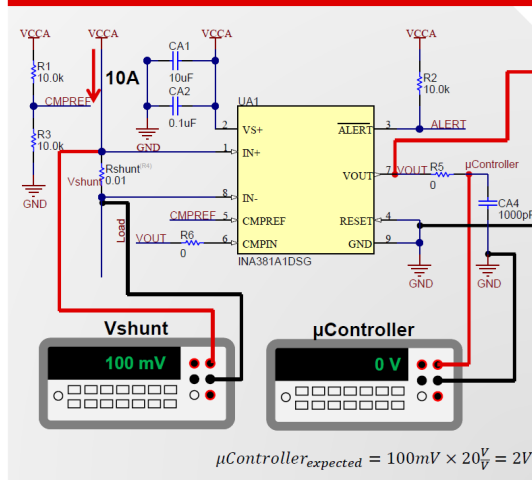
- But when measure the TI chip's output through a 0Ω resistor R_5 , we got $0V$.

Question: Did we get a bad or fake TI part?



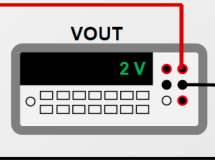
Example 1: TI Current Sense Amplifier

Conditions

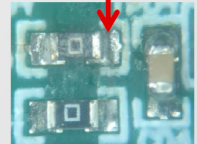


Source of Error

- Probing at the wrong spot



- Solder issue





Common Circuit Mistakes # 2: Ignoring Device Specifications

You may use quite a few electronic components in your circuits, from a commercial integrated circuits (IC, or chip) to transistors.

When things don't work, it is easy to blame that you had a bad part.

- Do you know most parts are more **robust** than you thought?
- Maybe you just didn't read how it is supposed to be used correctly.

Tip 2: Read the **specifications** in a datasheet.

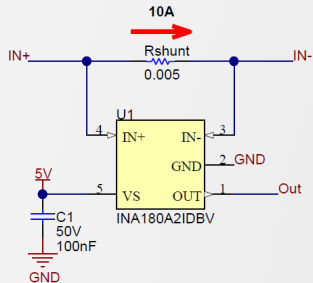


Example 2: Overlooking the Specification

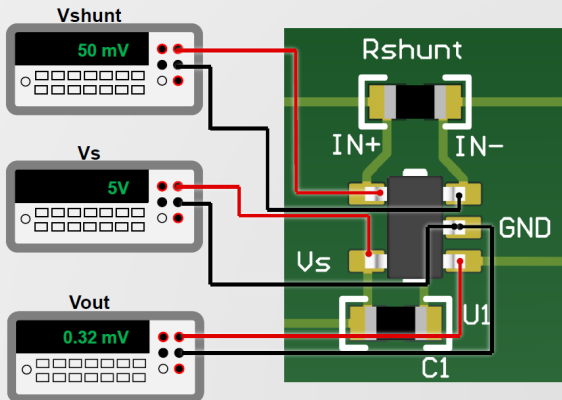
Let's look at another TI example. What's wrong with this board?

- The raw input $I \times R_{shunt} = 10A \times 5m\Omega = 50mV$ is correct.
- But V_{out} reads only 0.32mV instead of expected $100mV \cdot 50 = 2.5V$

Conditions



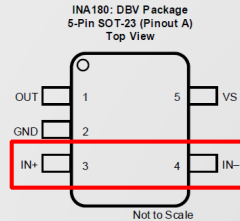
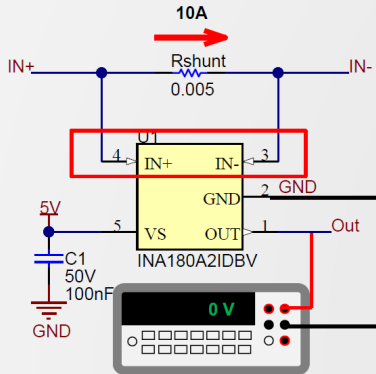
$$\text{Expected } V_{out} = 50mV \times 50 \frac{V}{V} = 2.5V$$





Example 2: Overlooking the Specification

Source of Error





Common Mistakes in this Lab

I am compiling a Root Cause Table (RCT) on our Forum. My message:

- Report more bugs!
- I can then compile a more useful RCT

Normal Topics

 Root Cause Table
Started by johnhu

0
Replies

4
Views

September 19, 2026, 09:50:30 pm
by johnhu



 Root Cause Table
« on: September 19, 2026, 09:50:30 pm »

 Quote  Modify  Remove

This is a working progress. I will start from Spring 2025 Bug tickets.

Symptom	Root Cause	Bug Ticket	Circuit	Contributor
Voltage amplitude on the oscilloscope is off	The probe (1x/10x) setting was different from the oscilloscope's setting.	2025-02-04	Any	Anonymous



References I

Ash, Andrew J. and John Hu (June 2025). "BOARD # 356: ECR: BCSEER: Are Females Better at Debugging Circuits?" In: *2025 ASEE Annual Conf. Proc.* URL: <https://peer.asee.org/board-356-ecr-bcser-are-females-better-at-debugging-circuits>.



Thank you!

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